



GIRRAWEE HIGH SCHOOL
MATHEMATICS EXTENSION 2

TASK 1 2013 December 2012: COMPLEX NUMBERS
ANSWERS COVER SHEET

Name: _____

Teacher: _____

**FINAL
MARK**

	MARK	E2	E3	E4	E5	E6	E7	E8
1 Multiple Choice	/5		✓					
2	/21		✓					
	/21							
3	/14		✓					
	/14							
4	/14		✓					
	/14							
5	/13		✓					
	/13							
6	/16		✓					
	/16							
7	/14		✓					
	14							
TOTAL	/97		/97					

HSC Outcomes

Mathematics Extension 2

- E1 appreciates the creativity, power and usefulness of mathematics to solve a broad range of problems.
- E2 chooses appropriate strategies to construct arguments and proofs in both concrete and abstract settings.
- E3 uses the relationship between algebraic and geometric representations of complex numbers and of conic sections.
- E4 uses efficient techniques for the algebraic manipulation required in dealing with questions such as those involving conic sections and polynomials.
- E5 uses ideas and techniques from calculus to solve problems in mechanics involving resolution of forces, resisted motion and circular motion
- E6 combines the ideas of algebra and calculus to determine the important features of the graphs of a wide variety of functions.
- E7 uses the techniques of slicing and cylindrical shells to determine volumes.
- E8 applies further techniques of integration, including partial fractions, integration by parts and recurrence formulae, to problems.
- E9 communicates abstract ideas and relationships using appropriate notation and logical argument.



GIRRAWEE HIGH SCHOOL

TASK 1 2013 (December 2012)

MATHEMATICS

EXTENSION 2

Complex Numbers

Time allowed – 100 Minutes

DIRECTIONS TO CANDIDATES

- Attempt ALL questions.
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Board-approved calculators may be used.
- Each question attempted is to be returned on a *separate* piece of paper clearly marked Question 1 , Question 2 , etc. Each piece of paper must show your name.
- You may ask for extra pieces of paper if you need them.
- For Multiple choice: Circle the correct answer on your examination paper.

Question 1 (Multiple Choice) Circle the correct answer on the Examination Paper

(a) The value of i^{27} is:

- (A) i (B) -1 (C) $-i$ (D) 1

(b) If $z = 2 + i$ and $w = 3 - 2i$ then $z\bar{w} =$

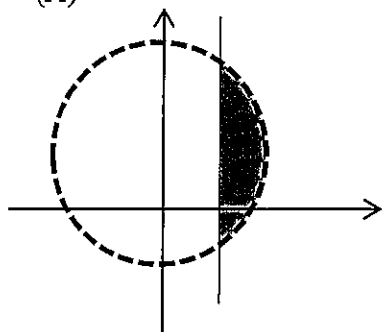
- (A) $8 + 7i$ (B) $4 + 7i$ (C) $8 - 7i$ (D) $4 - 7i$

(c) When expressed in modulus/ argument form, $3 - i\sqrt{3} =$

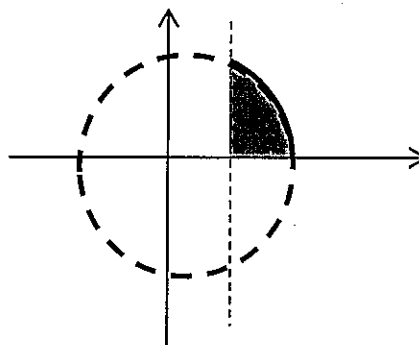
- (A) $2\sqrt{3}(\cos \frac{-\pi}{6} + i\sin \frac{-\pi}{6})$ (B) $2\sqrt{3}(\cos \frac{-\pi}{3} + i\sin \frac{-\pi}{3})$ (C) $2\sqrt{3}(\cos \frac{-5\pi}{6} + i\sin \frac{-5\pi}{6})$
 (D) $2\sqrt{3}(\cos \frac{-2\pi}{3} + i\sin \frac{-2\pi}{3})$

(d) The region in the complex plane defined by $|z - i| \leq 2$ and $\text{Re}(z) > 1$ is represented by:

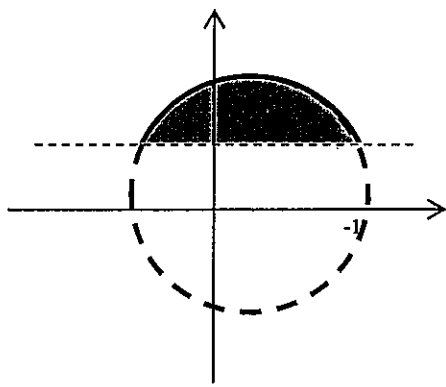
(A)



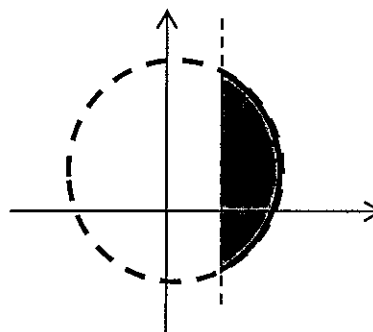
(B)



(C)



(D)



(e) In Cartesian form, $\frac{11-7i}{3-5i} =$

- (A) $2 - 76i$ (B) $2 + i$ (C) $\frac{2-76i}{34}$ (D) $\frac{2+i}{34}$

For Question 2 onward show all workings on the blank paper provided:

Question 2 (21 Marks)

Marks

- (a) (i) Find $\frac{-1+i}{1+i\sqrt{3}}$ in Cartesian form. 2
- (ii) Convert both $-1 + i$ and $1 + i\sqrt{3}$ to Modulus/argument form. 4
- (iii) Using the answers to (i) and (ii) find the value of $\cos \frac{5\pi}{12}$. 2
- (b) (i) If $x + iy = \sqrt{5 - 12i}$ find the exact value of x and y . 5
- (ii) Hence solve the equation $z^2 - 5z + (5 + 3i) = 0$ 3
- (c) Use DeMoivre's theorem to find $(-1 + i\sqrt{3})^5$ in Cartesian form. 3
- (d) Find all three cube roots of $-4\sqrt{2} + 4i\sqrt{2}$. Leave your answers in modulus/ argument form. 2

Question 3 (14 Marks)

- (a) Sketch each of the following loci on separate Argand diagrams:
- (i) $z\bar{z} = 2 \times \text{Re}(z)$ 3
- (ii) $\text{Arg}(z + 2 - i) = -\frac{\pi}{4}$ 2
- (iii) $|z + 1 - 2i| = |z - 1|$ 2
- (iv) $|z - i| = \text{Im}(z + i)$ 3
- (b) Sketch and shade the region satisfied by $|z - 2i| \leq 2$ and $\frac{\pi}{4} < \text{Arg } z \leq \frac{\pi}{2}$ on an Argand diagram. 4

Examination continues on the next page

Question 4 (14 Marks)

Marks

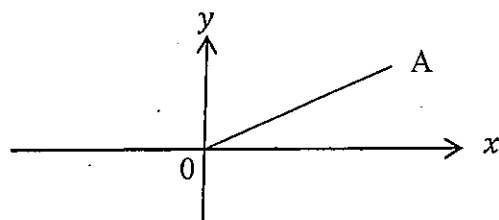
(a) z is an arbitrary point on the Argand diagram such that $\overrightarrow{OA} = z$.

(see diagram below). Copy the diagram on to your writing paper and draw in

(i) $\overrightarrow{OB}$ so that $\overrightarrow{OB} = iz$ 1

(ii) $\overrightarrow{OC}$ so that $\overrightarrow{OC} = z \times (\cos \frac{-\pi}{3} + i \sin \frac{-\pi}{3})$ 1

(iii) $\overrightarrow{OD}$ so that $\overrightarrow{OD} = -z$ 1

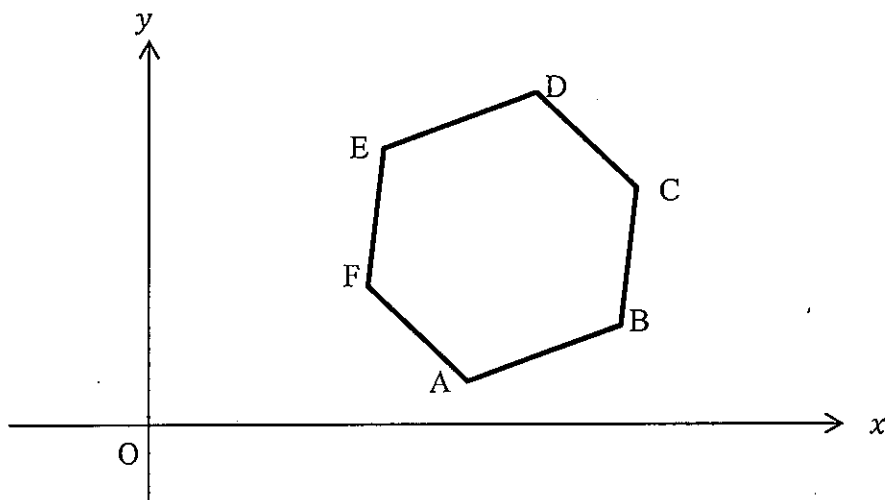


(b) On the diagram below, ABCDEF is a regular hexagon. If $\overrightarrow{OA} = z_1$ and

$$\overrightarrow{OB} = z_2$$

(i) Express $\overrightarrow{AB}$, $\overrightarrow{BC}$ and $\overrightarrow{AD}$ in terms of z_1 and z_2 . 3

(ii) Show that $\overrightarrow{AB} \times \overrightarrow{BC} \times \overrightarrow{CD} = (z_1 - z_2)^3$ 2



Question 4 continues on the next page

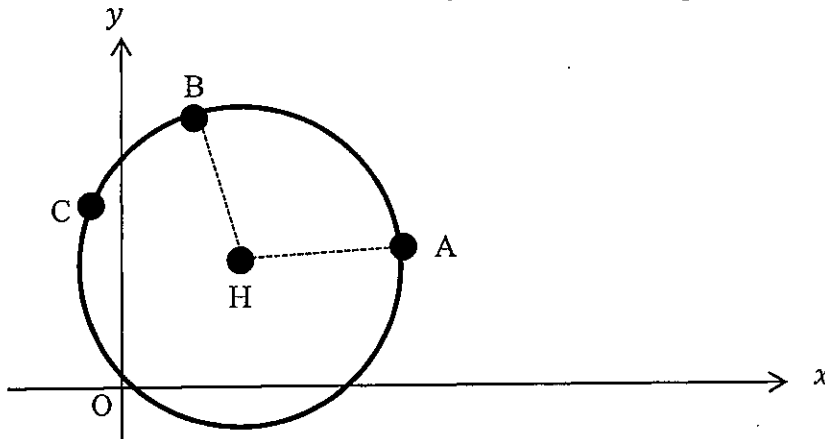
Question 4 (continued)

Marks

(c) On the diagram below, A, O and B are at the circumference of a circle with centre H. If $\overrightarrow{OA} = z_1$, $\overrightarrow{OH} = z_2$ and $\overrightarrow{OB} = z_3$

(i) Show that $\text{Arg} \left\{ \frac{z_3 - z_2}{z_1 - z_2} \right\} = \text{Arg} \left\{ \frac{z_3}{z_1} \right\}^2$ 4

(ii) C is a point on the circle in the major arc AB. If $\overrightarrow{OC} = z_4$ 2
show that $\text{Arg} \left\{ 1 - \frac{z_1}{z_4} \right\} = \text{Arg} \left\{ 1 - \frac{z_1}{z_3} \right\}$



Question 5 (13 Marks)

(a) The graph of $\text{Arg} \left(\frac{z+2}{z-4} \right) = \frac{\pi}{3}$ represents part of a circle. *Note that this is not a semicircle.*

Draw the graph of the part of the circle represented by $\text{Arg} \left(\frac{z+2}{z-4} \right) = \frac{\pi}{3}$ 6
and find the centre and radius of the circle and its Cartesian equation.

(b) (i) If $z = \cos \theta + i \sin \theta$ show that $z - \frac{1}{z} = 2i \sin \theta$ and that 4

$$z^n - \frac{1}{z^n} = 2i \sin n\theta.$$

(i) Hence express $\sin^5 \theta$ in terms of $\sin 5\theta$, $\sin 3\theta$ and $\sin \theta$. 3

Question 6 (16 Marks)**Marks**

- (a) (i) Use DeMoivre's Theorem to find the formula for $\cos 6\theta$ 2
(You may leave your answer in terms of mixtures of $\cos\theta$ and $\sin\theta$.)
- (ii) Hence show that 2
$$\cos 6\theta = (\cos^2\theta - \sin^2\theta)(\cos^4\theta - 14\cos^2\theta\sin^2\theta + \sin^4\theta)$$
- (iii) Hence using that $\cos \frac{\pi}{2} = 0$ show that $\cos \frac{\pi}{12}$ is a root of the 2
equation $16x^4 - 16x^2 + 1 = 0$
- (iv) Hence find the exact value of $\cos \frac{\pi}{12}$. 3
- (b)(i) Resolve $z^6 + 1$ into real quadratic factors. 3
- (ii) Hence show that 4
$$\cos 3\theta = 4\cos\theta(\cos^2\theta - \cos^2\frac{\pi}{6})$$

Question 7 (14 Marks)

- (a) (i) Use that $z^{10} - 1 = (z^2)^5 - 1$ to factorise $z^{10} - 1$. 1
- (ii) Hence or otherwise find the roots of the equation 1
$$z^8 + z^6 + z^4 + z^2 + 1 = 0.$$
- (iii) Hence show that $\cos \frac{\pi}{5}$ is a root of the equation 2
$$2\cos 4\theta + 2\cos 2\theta + 1 = 0.$$

(You may assume that if $z = \cos\theta + i\sin\theta$ then $z^n + \frac{1}{z^n} = 2\cos n\theta$)
- (iv) Hence find the exact value of $\cos \frac{\pi}{5}$. 4
- (b) Let w be the root of $z^{10} - 1 = 0$ with the smallest positive argument.
- (i) Show that $w^2, w^3, w^4, w^5, w^6, w^7, w^8$ and w^9 are the other 1
non real roots of $z^{10} - 1 = 0$.
- (ii) By expanding $(w + w^9)(w^2 + w^8)(w^3 + w^7)(w^4 + w^6)$ 5
and using that w is a root of $z^8 + z^6 + z^4 + z^2 + 1 = 0$ show that
$$16\cos^2\frac{\pi}{5}\cos^2\frac{2\pi}{5} = 1.$$

End of examination

Solutions to Complex Numbers test p. 1
Dec. 2012 for 2013 MSC

(1) (a) C (b) B (c) A (d) D (e) B

Q.2 = (21)

$$(2) (a) (i) \frac{-1+i}{1+i\sqrt{3}} \times \frac{(-1-i\sqrt{3})}{(-1-i\sqrt{3})} \quad |$$

$$= \frac{(\sqrt{3}-1) + i(\sqrt{3}+1)}{4} \quad |$$

$$(ii) -1+i = \sqrt{2} \left(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4} \right) \quad |$$

$$1+i\sqrt{3} = 2 \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right) \quad |$$

(iii) Using mod/arg form,

$$\frac{-1+i}{1+i\sqrt{3}} = \frac{1}{\sqrt{2}} \cos \frac{5\pi}{12} + i \frac{1}{\sqrt{2}} \sin \frac{5\pi}{12} \quad |$$

Using (i) & equating reals:

$$\frac{\sqrt{3}-1}{4} = \frac{1}{\sqrt{2}} \cos \frac{5\pi}{12} \quad |$$

$$\frac{\sqrt{6}-\sqrt{2}}{4} = \cos \frac{5\pi}{12} \quad |$$

(b) $(x+iy)^2 = 5-12i$ x, y real.

$$x^2 - y^2 + 2ixy = 5 - 12i \quad |$$

equating reals: $x^2 - y^2 = 5$ (1)

equating imaginaries: $2xy = -12$

$$y = -\frac{6}{x} \quad (2) \quad |$$

Sub (2) in (1): $x^2 - \left(-\frac{6}{x}\right)^2 = 5$ $|$

$$x^2 - \frac{36}{x^2} = 5$$

PTO $\rightarrow$

Q. (2) (b) (i) [continued]:

p. 2

$$x^4 - 36 = 5x^2$$

$$x^4 - 5x^2 - 36 = 0$$

$$(x^2 - 9)(x^2 + 4) = 0$$

$$x^2 = 9 \quad [\text{as } x \text{ is real, } x^2 \neq -4]$$

$$x = \pm 3.$$

$$\text{As } y = -\frac{6}{x}, \quad y = \mp 2, \quad | \quad \underline{5}$$

$$\sqrt{5-12i} = \pm (3-2i).$$

$$(ii) \quad z^2 - 5z + (5+3i) = 0$$

$$z = \frac{5 \pm \sqrt{(-5)^2 - 4 \times 1 \times (5+3i)}}{2} \quad |$$

$$= \frac{5 \pm \sqrt{5-12i}}{2}$$

$$= \frac{5 \pm (3-2i)}{2} \quad [\text{using (i)}] \quad |$$

$$z = \frac{5+(3-2i)}{2} \quad \text{or} \quad z = \frac{5-(3-2i)}{2} \quad \underline{3}$$

$$z = 4-i \quad \text{or} \quad z = 1+i \quad |$$

$$(c) \quad (-1+i\sqrt{3})^5$$

$$= 2 \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right)^5 \quad |$$

$$= 32 \left(\cos \frac{10\pi}{3} + i \sin \frac{10\pi}{3} \right) \quad |$$

$$= 32 \left(\cos -\frac{2\pi}{3} + i \sin -\frac{2\pi}{3} \right) \quad \underline{3}$$

$$= 32 \left(-\frac{1}{2} - i \frac{\sqrt{3}}{2} \right)$$

$$= -16 - 16i\sqrt{3} \quad |$$

Solutions to Complex N^{os} p.3

Q.(2)(d) $-4\sqrt{2} + 4i\sqrt{2}$

$= 64 \left(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4} \right)$

$\therefore \sqrt[3]{-4\sqrt{2} + 4i\sqrt{2}}$

$= 4 \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right) + \frac{2k\pi}{3}, k=0,1,2$

$= 4 \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right), 4 \left(\cos \frac{11\pi}{12} + i \sin \frac{11\pi}{12} \right), 4 \left(\cos \frac{19\pi}{12} + i \sin \frac{19\pi}{12} \right)$

$\rightarrow$ can also be

$4 \left(\cos \frac{-5\pi}{12} + i \sin \frac{-5\pi}{12} \right)$

Q.(3)(a)(i) Let $z = x + iy$

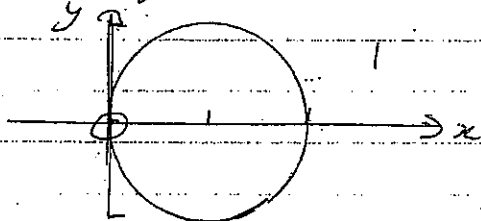
$z\bar{z} = 2 \times \text{Re}(z)$

$(x+iy)(x-iy) = 2x$

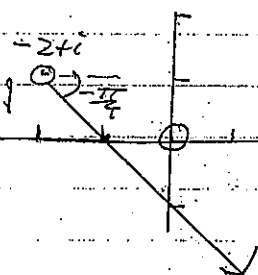
$x^2 + y^2 = 2x$

$(x^2 - 2x + 1) + y^2 = 1$

$(x-1)^2 + y^2 = 1$



(ii) $\text{Arg}(z + 2 - i) = -\frac{\pi}{4}$



(iii) $|z + 1 - 2i| = |z - 1|$

$\sqrt{(x+1)^2 + (y-2)^2} = \sqrt{(x-1)^2 + y^2}$

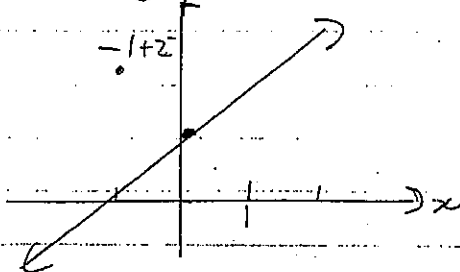
$x^2 + 2x + 1 + y^2 - 4y + 4 = x^2 - 2x + 1 + y^2$

$4x + 4 = 4y$

$x + 1 = y$ or $y = x + 1$

[Note: We could also perpendicular bisector of line between

$-1 + 2i$ & 1 .



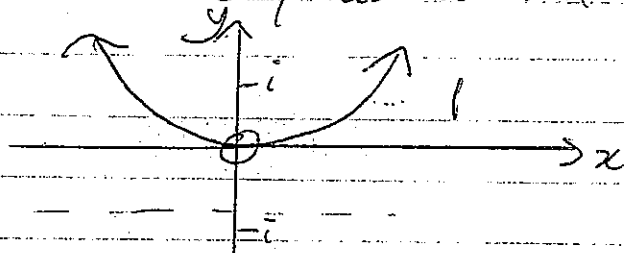
Q. (3)(a)(iv) $|z-i| = \text{Im}(z+i)$

$$\sqrt{x^2 + (y-1)^2} = y+1$$

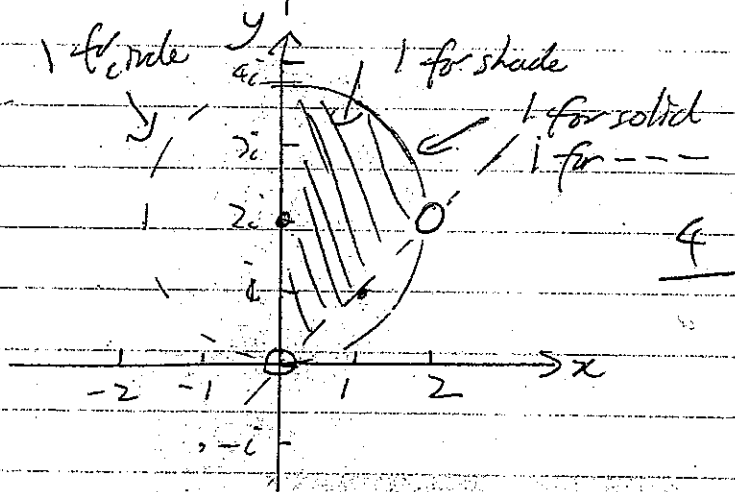
$$x^2 + y^2 - 2y + 1 = y^2 + 2y + 1$$

$$x^2 = 4y$$

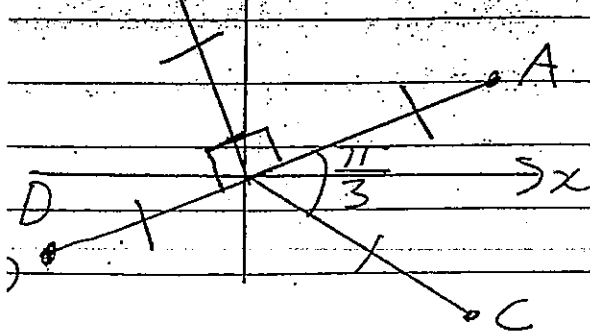
Note: We could also have noted that this was a parabola with focus at i & directrix at $y = -1$.
[so total $\text{Im} = -i$]



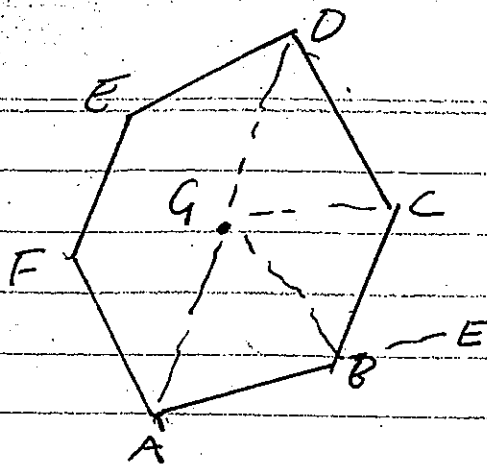
(6)



Q. (4)(a)



(4)(b)



p.5 Solutions
2013 MSc
[Dec 2012]
Complex NOS

$$(i) \vec{AB} = z_2 - z_1$$

$\vec{BC}$: Let $\vec{BE}$ continue $\vec{AB}$

[be $\vec{AB}$ produced]

$\angle CBE = \frac{\pi}{3}$ [exterior $\angle$ regular hexagon]

$$\therefore \vec{BC} = \text{cis } \frac{\pi}{3} \times \vec{AB}$$

$$= \text{cis } \frac{\pi}{3} (z_2 - z_1)$$

$\vec{AD}$: Let G be midpoint of AD.

As AG bisects $\angle FAB$ & BG bisects $\angle ABC$

$$\angle GAB = \angle GBA = \frac{\pi}{3}$$

$\therefore \triangle AGB$ is equilateral (& $AB = BG = AG$).

$$\therefore \vec{AG} = \text{cis } \frac{\pi}{3} \times \vec{AB}$$

$$\vec{AD} = 2 \text{cis } \frac{\pi}{3} \times \vec{AB}$$

$$= 2 \text{cis } \frac{\pi}{3} (z_2 - z_1)$$

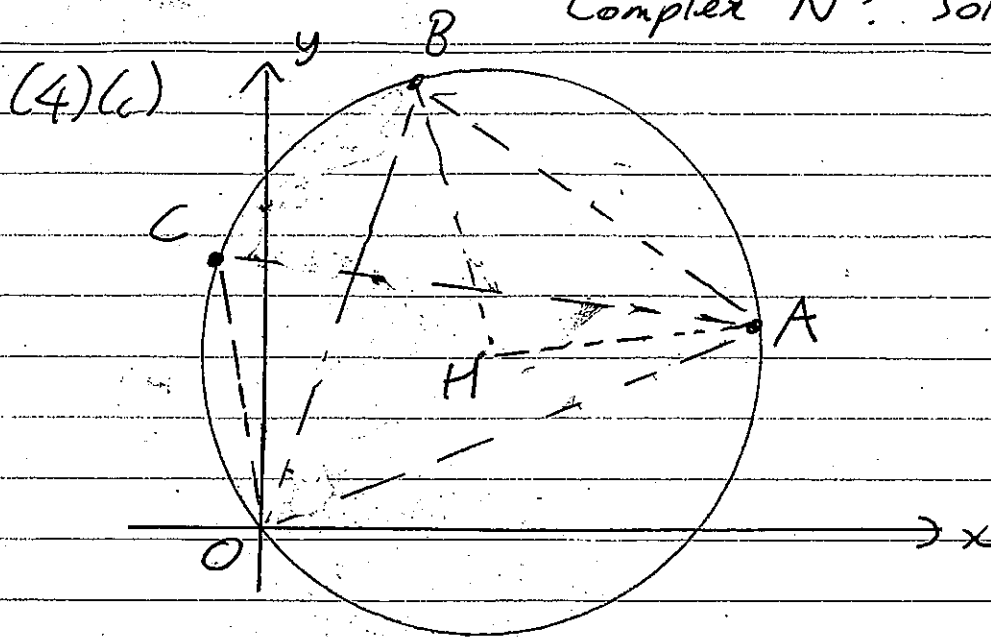
$$(ii) \vec{AB} \times \vec{BC} \times \vec{CD}$$

$$= (z_2 - z_1) \times \text{cis } \frac{\pi}{3} (z_2 - z_1) \times \text{cis } \frac{2\pi}{3} (z_2 - z_1)$$

$$= \text{cis } \pi \times (z_2 - z_1)^3$$

$$= - (z_2 - z_1)^3$$

$$= (z_1 - z_2)^3$$



$$(i) \begin{aligned} z_3 - z_2 &= \vec{AB} \\ z_1 - z_2 &= \vec{HA} \end{aligned}$$

$$\angle BHA = \text{Arg}(z_3 - z_2) - \text{Arg}(z_1 - z_2)$$

$$\angle BHA = \text{Arg}\left(\frac{z_3 - z_2}{z_1 - z_2}\right)$$

$$\text{Similarly as } \vec{OB} = z_3, \vec{OA} = z_1$$

$$\angle BOA = \text{Arg}\left(\frac{z_3}{z_1}\right)$$

As $\angle BOA = 2 \times \angle BHA$ [Lat centre = $2 \times$ Lat circumference on same arc].

$$\text{Arg}\left(\frac{z_3 - z_2}{z_1 - z_2}\right) = 2 \text{Arg}\left(\frac{z_3}{z_1}\right)$$

$$\text{Arg}\left(\frac{z_3 - z_2}{z_1 - z_2}\right) = \text{Arg}\left[\left(\frac{z_3}{z_1}\right)^2\right] \text{ [by De Moivre].}$$

(ii)

$$\therefore \angle OCA = \text{Arg}\left(\frac{\vec{AC}}{\vec{OC}}\right)$$

$$= \text{Arg}\left(\frac{z_4 - z_1}{z_4}\right)$$

$$\vec{AB} = z_3 - z_1$$

$$\therefore \angle OBA = \text{Arg}\left(\frac{z_3 - z_1}{z_3}\right)$$

PTO $\Rightarrow$

Q. (5)(b)(i) $z = \cos \theta + i \sin \theta$

$$\therefore \frac{1}{z} = \cos(-\theta) + i \sin(-\theta) \quad [\text{by De Moivre}]$$

$$= \cos \theta - i \sin \theta \quad [\text{as } \cos \text{ even, } \sin \text{ odd}]$$

$$\therefore z - \frac{1}{z} = (\cos \theta + i \sin \theta) - (\cos \theta - i \sin \theta) = 2i \sin \theta.$$

Similarly $z^n - \frac{1}{z^n} = \cos n\theta + i \sin n\theta - [\cos(-n\theta) + i \sin(-n\theta)]$
 $\quad \quad \quad [\text{by De Moivre}]$
 $\quad \quad \quad = \cos n\theta + i \sin n\theta - [\cos n\theta - i \sin n\theta]$
 $\quad \quad \quad [\text{as } \cos \text{ even, } \sin \text{ odd}]$
 $\quad \quad \quad = 2i \sin n\theta.$

(ii) Hence if $z = \cos \theta + i \sin \theta$,

$$\left(z - \frac{1}{z}\right)^5 = z^5 - 5z^3 + 10z - \frac{10}{z} + \frac{5}{z^3} - \frac{1}{z^5}$$

$$(2i \sin \theta)^5 = \left(z^5 - \frac{1}{z^5}\right) - 5\left(z^3 - \frac{1}{z^3}\right) + 10\left(z - \frac{1}{z}\right)$$

$$32i \sin^5 \theta = 2i \sin 5\theta - 10i \sin 3\theta + 20i \sin \theta \quad [\text{using (i)}]$$

$$\div \text{BS by } 32i$$

$$\sin^5 \theta = \frac{\sin 5\theta}{16} - \frac{5 \sin 3\theta}{16} + \frac{10 \sin \theta}{16}$$

$$\sin^5 \theta = \frac{1}{16} [\sin 5\theta - 5 \sin 3\theta + 10 \sin \theta]$$

Q. (6) By De Moivre (a) (i)

$$(\cos \theta + i \sin \theta)^6$$

$$= \cos 6\theta + i \sin 6\theta$$

$$\cos^6 \theta + 6i \cos^5 \theta \sin \theta - 15 \cos^4 \theta \sin^2 \theta - 20i \cos^3 \theta \sin^3 \theta$$

1

$$+ 15 \cos^2 \theta \sin^4 \theta + 6i \cos \theta \sin^5 \theta - \sin^6 \theta$$

$$= \cos 6\theta + i \sin 6\theta$$

Equating real parts

1 2

$$\text{Hence } \cos 6\theta = \cos^6 \theta - 15 \cos^4 \theta \sin^2 \theta + 15 \cos^2 \theta \sin^4 \theta - \sin^6 \theta$$

$$\begin{aligned} \text{(ii)} \quad \cos 6\theta &= \cos^6 \theta - \sin^6 \theta - 15 \cos^2 \theta \sin^2 \theta (\cos^2 \theta - \sin^2 \theta) \\ &= (\cos^2 \theta - \sin^2 \theta) (\cos^4 \theta + \cos^2 \theta \sin^2 \theta + \sin^4 \theta) - 15 \cos^2 \theta \sin^2 \theta (\cos^2 \theta - \sin^2 \theta) \end{aligned}$$

$$= (\cos^2 \theta - \sin^2 \theta) (\cos^4 \theta - 14 \cos^2 \theta \sin^2 \theta + \sin^4 \theta) \quad \underline{1}$$

$$\text{(iii)} \quad \text{As } \cos \frac{\pi}{2} = 0$$

$$\cos \frac{\pi}{2} \text{ is a solution to } \cos 6\theta = 0$$

$$\text{As } \cos \frac{2\pi}{12} \neq \sin^2 \frac{\pi}{2},$$

$$\cos \frac{\pi}{2} \text{ must be a solution to}$$

$$\cos^4 \theta - 14 \sin^2 \theta \cos^2 \theta + \sin^4 \theta = 0 \quad [\text{from (ii)}] \quad \underline{1}$$

$$\cos^4 \theta - 14 \cos^2 \theta (1 - \cos^2 \theta) + (1 - \cos^2 \theta)^2 = 0$$

$$\cos^4 \theta - 14 \cos^2 \theta + 14 \cos^4 \theta + 1 - 2 \cos^2 \theta + \cos^4 \theta = 0$$

$$16 \cos^4 \theta - 16 \cos^2 \theta + 1 = 0 \quad \underline{2}$$

$$\text{(iv)} \quad \cos^2 \frac{\pi}{12} = \frac{16 \pm \sqrt{16^2 - 4 \times 16 \times 1}}{2 \times 16}$$

$$= \frac{16 + 8\sqrt{3}}{32} \quad \text{or} \quad \frac{16 - 8\sqrt{3}}{32}$$

$$\cos^2 \frac{\pi}{12} = \frac{2 + \sqrt{3}}{4} \quad \text{or} \quad \cos^2 \frac{\pi}{12} = \frac{2 - \sqrt{3}}{4} \quad \underline{1}$$

$$\text{Note: As } \cos \frac{\pi}{12} > \cos \frac{\pi}{6} \text{ \& } \cos \frac{2\pi}{6} = \frac{3}{4},$$

$$\cos^2 \frac{\pi}{12} = \frac{2 + \sqrt{3}}{4} \quad \underline{1}$$

$$\cos \frac{\pi}{12} = \frac{\sqrt{2 + \sqrt{3}}}{2} \quad \underline{1}$$

Q.6(b)(i) Roots of $z^6 + 1 = 0$ are

$$z = \cos \frac{\pi}{6} + i \sin \frac{\pi}{6}, \cos \frac{2\pi}{6} + i \sin \frac{2\pi}{6}, \cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6}, \cos \frac{7\pi}{6} + i \sin \frac{7\pi}{6}, \cos \frac{3\pi}{6} + i \sin \frac{3\pi}{6}, \cos \frac{4\pi}{6} + i \sin \frac{4\pi}{6}$$

$$\text{Hence } z^6 + 1 = \left(z - \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right) \right) \left(z - \left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6} \right) \right) (z - i)(z + i) \left(z - \left(\cos \frac{3\pi}{6} + i \sin \frac{3\pi}{6} \right) \right) \left(z - \left(\cos \frac{7\pi}{6} + i \sin \frac{7\pi}{6} \right) \right)$$

Noting that $\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6} = \cos \frac{\pi}{6} + i \sin \frac{\pi}{6}$
 $= \cos \frac{\pi}{6} - i \sin \frac{\pi}{6}$ (as cos even, sin odd).

Similarly $\cos \frac{7\pi}{6} + i \sin \frac{7\pi}{6} = \cos \frac{\pi}{6} - i \sin \frac{\pi}{6}$.

$$z^6 + 1 = \left(z^2 - 2z \cos \frac{\pi}{6} + 1 \right) (z^2 + 1) \left(z^2 - 2z \cos \frac{\pi}{6} + 1 \right) \quad \text{* Note: 1}$$

As $\cos \frac{5\pi}{6} = -\cos \frac{\pi}{6}$
 $\rightarrow$ Full marks for (i) to be.

$$z^6 + 1 = \left(z^2 - 2z \cos \frac{\pi}{6} + 1 \right) (z^2 + 2z \cos \frac{\pi}{6} + 1) (z^2 + 1)$$

$$\stackrel{\div z^3}{\Rightarrow} z^3 + \frac{1}{z^3} = \left(z + \frac{1}{z} - 2 \cos \frac{\pi}{6} \right) \left(z + \frac{1}{z} + 2 \cos \frac{\pi}{6} \right) (z + \frac{1}{z})$$

If $z = \cos \theta + i \sin \theta$,

$$z^n + \frac{1}{z^n} = \cos n\theta + i \sin n\theta + \cos(-n\theta) + i \sin(-n\theta) \quad \text{[by De Moivre]}$$

$$= \cos n\theta + i \sin n\theta + \cos n\theta - i \sin n\theta \quad \text{(as cos even, sin odd)}$$

$$= 2 \cos n\theta$$

$$2 \cos 3\theta = \left(2 \cos \theta - 2 \cos \frac{\pi}{6} \right) \left(2 \cos \theta + 2 \cos \frac{\pi}{6} \right) 2 \cos \theta$$

$$\cos 3\theta = 4 \cos \theta \left(\cos \theta - \cos \frac{\pi}{6} \right) \left(\cos \theta + \cos \frac{\pi}{6} \right)$$

$$\cos 3\theta = 4 \cos \theta \left(\cos^2 \theta - \cos^2 \frac{\pi}{6} \right)$$

as required.

P. 11

$$(7)(a)(i) z^{10} - 1 = (z^2)^5 - 1$$

$$= (z^2 - 1)(z^8 + z^6 + z^4 + z^2 + 1) \quad \underline{1}$$

(ii) Hence roots of $z^8 + z^6 + z^4 + z^2 + 1 = 0$ are

$$z = \cos \frac{\pi}{5} + i \sin \frac{\pi}{5}, z = \cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5}, z = \cos \frac{3\pi}{5} + i \sin \frac{3\pi}{5},$$

$$z = \cos \frac{4\pi}{5} + i \sin \frac{4\pi}{5}, z = \cos \frac{6\pi}{5} + i \sin \frac{6\pi}{5}, z = \cos \frac{7\pi}{5} + i \sin \frac{7\pi}{5},$$

$$z = \cos \frac{8\pi}{5} + i \sin \frac{8\pi}{5}, z = \cos \frac{9\pi}{5} + i \sin \frac{9\pi}{5}. \quad \underline{1}$$

(iii) $z^8 + z^6 + z^4 + z^2 + 1 = 0$

÷ by z^4 & rearranging

$$\left(z^4 + \frac{1}{z^4}\right) + \left(z^2 + \frac{1}{z^2}\right) + 1 = 0 \quad \underline{1}$$

As $z = \cos \theta + i \sin \theta, z^n + \frac{1}{z^n} = 2 \cos n\theta$

$$2 \cos 4\theta + 2 \cos 2\theta + 1 = 0 \text{ if } \theta = \frac{\pi}{5}, \frac{2\pi}{5} \text{ etc.} \quad \underline{2}$$

$\therefore 2 \cos 4\theta + 2 \cos 2\theta + 1 = 0$ has $\cos \frac{\pi}{5}$ as a root. 1

(iv) Could be done using $4\theta = 2 \times 2\theta$

Hence $\cos 4\theta = 2 \cos^2 2\theta - 1$ [See next page] $\rightarrow$

& solving for $\cos \frac{2\pi}{5}$. then using $\cos \frac{2\pi}{5} = 2 \cos^2 \frac{\pi}{5} - 1$.

However, $\cos \frac{4\pi}{5} = -\cos \frac{\pi}{5}$.

So as $2 \cos \frac{4\pi}{5} + 2 \cos \frac{2\pi}{5} + 1 = 0$

$$-2 \cos \frac{\pi}{5} + 2(2 \cos^2 \frac{\pi}{5} - 1) + 1 = 0 \quad \underline{1}$$

$$4 \cos^2 \frac{\pi}{5} - 2 \cos \frac{\pi}{5} - 1 = 0 \quad \underline{1}$$

$$\cos \frac{\pi}{5} = \frac{2 \pm \sqrt{2^2 - 4 \times 4 \times (-1)}}{2 \times 4} \quad \underline{4}$$

$$= \frac{2 \pm 2\sqrt{5}}{8}$$

$$\cos \frac{\pi}{5} = \frac{1 \pm \sqrt{5}}{4} \quad \underline{1}$$

As $\frac{\pi}{5}$ is in Q_1 , $\cos \frac{\pi}{5}$ is positive.

$$\cos \frac{\pi}{5} = \frac{1 + \sqrt{5}}{4} \quad \underline{4}$$

Q. (7)(a)(iv) alternative:

$$2\cos 4\theta + 2\cos 2\theta + 1 = 0$$

$$\text{As } 4\theta = 2 \times 2\theta$$

$$\cos 4\theta = 2\cos^2(2\theta) - 1$$

Hence equation becomes:

$$2[2\cos^2 2\theta - 1] + 2\cos 2\theta + 1 = 0$$

$$4\cos^2 2\theta + 2\cos 2\theta - 1 = 0$$

$$\text{Hence } \cos 2\theta = \frac{-2 \pm \sqrt{2^2 - 4 \times 4 \times -1}}{2 \times 4}$$

$$\cos 2\theta = \frac{-1 \pm \sqrt{5}}{4}$$

As we are looking for $\cos \frac{2\pi}{5}$ at this stage, $\cos \theta$ must be positive.

$$\text{Hence } \cos \frac{2\pi}{5} = \frac{-1 + \sqrt{5}}{4}$$

$$\begin{aligned} \text{As } \cos \frac{2\pi}{5} &= \cos \left[2 \times \frac{\pi}{5} \right] \\ &= 2\cos^2 \left(\frac{\pi}{5} \right) - 1 \end{aligned}$$

$$2\cos^2 \frac{\pi}{5} - 1 = \frac{-1 + \sqrt{5}}{4}$$

$$\cos^2 \frac{\pi}{5} = \frac{3 + \sqrt{5}}{8}$$

$$\cos \frac{\pi}{5} = \sqrt{\frac{3 + \sqrt{5}}{8}} \quad \left[\text{As } \frac{\pi}{5} \text{ is in } Q_1, \right.$$

$\cos \frac{\pi}{5}$ is negative].

$$\text{To show } \sqrt{\frac{3 + \sqrt{5}}{8}} = \frac{1 + \sqrt{5}}{4}$$

$$\left(\frac{1 + \sqrt{5}}{4} \right)^2 = \frac{1 + 2\sqrt{5} + 5}{16} = \frac{6 + 2\sqrt{5}}{16} = \frac{3 + \sqrt{5}}{8}$$

Either $\sqrt{\frac{3 + \sqrt{5}}{8}}$ or $\cos \frac{\pi}{5} = \frac{1 + \sqrt{5}}{4}$ are acceptable.

Q. (7)(b) As $w = \cos \frac{\pi}{5} + i \sin \frac{\pi}{5}$.
(i.)

w^5, w^{10} are REAL roots.
Testing $w^2, w^3, w^4, w^6, w^7, w^8, w^9$
 $(w^2)^{10} = w^{20} = 1$

$$(w^3)^{10} = w^{30} = 1$$

$$(w^4)^{10} = w^{40} = 1$$

$$(w^6)^{10} = w^{60} = 1$$

$$(w^7)^{10} = w^{70} = 1$$

$$(w^8)^{10} = w^{80} = 1$$

$$(w^9)^{10} = w^{90} = 1$$

$w^2, w^3, w^4, w^6, w^7, w^8, w^9$ are other non real roots.

$$\begin{aligned} & (ii) (w + w^9)(w^2 + w^8)(w^3 + w^7)(w^4 + w^6) \\ &= (w^3 + w^9 + w^{11} + w^{17})(w^7 + w^9 + w^{11} + w^{13}) \\ &= w^{10} + w^{12} + w^{14} + w^{16} + w^{16} + w^{18} + w^{20} + w^{22} + w^{18} + w^{20} + w^{22} + w^{24} \\ &+ w^{26} + w^{28} + w^{30} \\ &= 1 + w^2 + w^4 + w^6 + w^8 + 1 + w^2 + w^8 + 1 + w^2 + w^4 \\ &+ w^4 + w^6 + w^8 + 1 \end{aligned}$$

$$= 4 + 3(w^2 + w^4 + w^6 + w^8)$$

$$= 1 + 3(1 + w^2 + w^4 + w^6 + w^8)$$

As $w = \cos \frac{\pi}{5} + i \sin \frac{\pi}{5}$ is a root of $z^8 + z^6 + z^4 + z^2 + 1 = 0$

$$1 + w^2 + w^4 + w^6 + w^8 = 0$$

$$\text{Hence } (w + w^9)(w^2 + w^8)(w^3 + w^7)(w^4 + w^6) = 1.$$

$$\text{Using } w + w^9 = 2\cos \frac{\pi}{5}, w^2 + w^8 = 2\cos \frac{2\pi}{5}, w^3 + w^7 = 2\cos \frac{3\pi}{5}, w^4 + w^6 = 2\cos \frac{4\pi}{5}$$

$$2\cos \frac{\pi}{5} \times 2\cos \frac{2\pi}{5} \times 2\cos \frac{3\pi}{5} \times 2\cos \frac{4\pi}{5} = 1$$

$$\text{As } \cos \frac{3\pi}{5} = -\cos \frac{2\pi}{5}, \cos \frac{4\pi}{5} = -\cos \frac{\pi}{5}$$

$$2\cos \frac{\pi}{5} \times 2\cos \frac{2\pi}{5} \times -2\cos \frac{2\pi}{5} \times -2\cos \frac{\pi}{5} = 1$$

$$16 \cos^2 \frac{\pi}{5} \cos^2 \frac{2\pi}{5} = 1.$$